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Dynamics of American Beech Regeneration 10 Years following Harvesting in a Beech Bark Disease-Affected Stand in Maine

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ABSTRACT: *Root sprouting following harvesting of American beech has resulted in the development of dense, slow-growing thickets in many stands throughout Maine and the northeastern United States. The problem is compounded by the presence of beech bark disease, as most sprouts arise from roots of disease-susceptible genotypes. Ten-year post-harvest regeneration conditions were examined in a central Maine beech stand harvested in both winter and summer with partial cutting and clearcuts. Initially, sprouts in winter cuts survived longer than those in summer cuts. After 10 years, we found that beech regeneration survival in the winter treatments continued to be higher than in the summer treatments. Disease-resistant American beech of high vigor left in clearcuts without the protection of surrounding trees were highly susceptible to decline and death from exposure. Leaving a small buffer of unharvested trees around resistant beech is recommended. North. J. Appl. For. 23(3):192–196.*

Key Words: American beech, beech bark disease, regeneration survival, root sprouting.

American beech in stands throughout Maine, the northeastern United States, and Maritime Canada have been damaged by the beech bark disease complex for well over 60 years (Houston 1994). Root systems of harvested beech, as well as those of uncut beech damaged by harvesting activities, often develop dense thickets of sprouts. These thickets grow slowly, occupying valuable stand space. Furthermore, the genotypes making up the beech thickets are composed mostly of beech susceptible to the beech bark disease. This has resulted in a serious vegetation management problem for land managers. Prediction and management of beech sprouting in uneven-aged stands has been recently investigated in New York (Bohn and Nyland 2003). Understanding the consequences of the season and intensity of harvest on root disturbance and the development and survival of beech resistant to the beech scale and hence the beech bark disease in even-aged systems is also key to improving the quality of affected stands. Determining the long-term effects of different harvesting practices will allow forest managers to evaluate the potential effects of harvesting on stands affected with beech bark disease and to make management decisions that will encourage the survival and propagation of resistant trees.

Houston (2001) examined beech regeneration in a stand harvested in both winter and summer, with partial cutting and clearcuts. He found that initially, sprouts in winter cuts survived longer than those in summer cuts. We revisited these plots 10 years after harvest and posed three questions: 1) Is there a difference in American beech sprout mortality between summer and winter cuts; 2) Is American beech regeneration survival greater around beech resistant to the beech scale than around susceptible beech; and 3) Do different harvesting levels affect the survival of residual resistant beech?

Study Area

The study area is located on the Maine Public Lands, Sebois Unit, in southern Piscataquis County in north central Maine. Established in 1989 by Houston (2001), the study area consisted of 15 10-ac blocks located on a hardwood ridge running from north to south and ranging in elevation from 495 to 760 ft. The forest in this area is primarily composed of American beech and includes sugar maple, yellow birch, red maple, striped maple, and white ash.

Five block-level seasonal harvest treatments were assigned to the 15 blocks, with each treatment replicated three times: summer clearcut blocks, summer partial-cut blocks, winter clearcut blocks, winter partial-cut blocks, and no-cut control blocks. Partial-cut treatments were in the form of an improvement harvest, where a priority was to remove low-quality stems. Some small (2–3-in. diameter) saplings were left uncut in the commercial clearcut treatments. Harvesting was done using chainsaws, with hand crews and skidders from January through March and June through August 1991. In 2002, the block boundaries were re-established using maps from the original study in combination with a compass and a GPS unit.

NOTE: William Ostrofsky can be reached at beechpath@yahoo.com. Amanda Farrar can be reached at manfarrar1974@yahoo.com. This report is based on an MS thesis completed by Amanda Farrar. This is publication number 2877 of the Maine Agricultural and Forest Experiment Station. Funding for this project was provided by the Maine Agricultural and Forest Experiment Station, University of Maine, Orono, ME.

Stand Plots: Overstory Data

After the original blocks were located, the original transect points were re-established. Transect points ran diagonally from corner to corner through each block (Figure 1). There were five transect points per block, for a total of 75. Data on overstory composition and structure were obtained using point samples centered on these points. Species, dbh, and crown class were recorded for trees selected using a BAF 10 prism. The estimated percentage of live crown, severity of scale infestation, presence of *Xylococculous betulae*, and severity of defect of all beech trees selected by the prism were also noted.

Stand Plots: Regeneration Data

Regeneration data were obtained using metric measurement units when the study was established (Houston 2001). Our remeasurements followed the protocol but have been converted to English units.

Data on regeneration were obtained in nested 3.28-ft (1-m) and 6.56-ft (2-m) radius plots located in the middle three of the five transect points in each block (Figure 1). The plot centers for each of the plots were located 40.5 ft from the transect points in the four cardinal directions. There were 12 of each type of plot in each block, for a total of 180 nested plots. In the 3.28-ft plots, all species <1 in. in diameter at 3.9 in. (<2.5 cm at 10 cm) from the ground were counted and recorded. Smaller regeneration was split into three height classes: <10, 10.1–19.68, and >19.68 in. (< 2.5, 25–50, and >50 cm). In the 6.56-ft radius plots, trees that were 1–3.9 in. in diameter at 3.9 in. from the ground and >3.28 ft tall were counted and sorted.

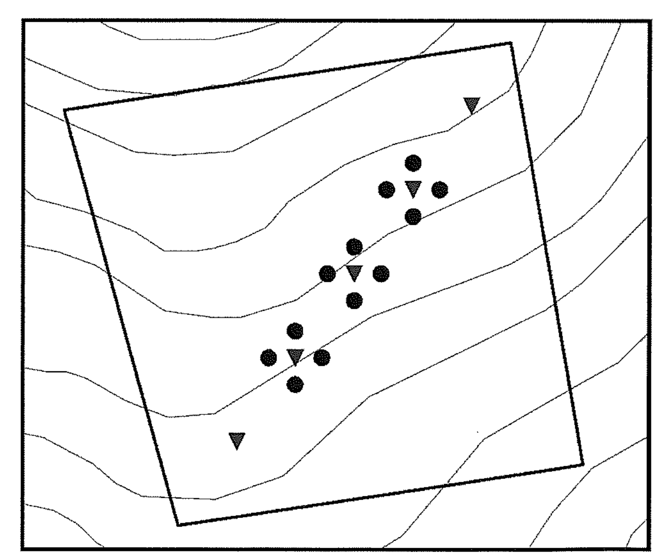

Figure 1. Sample nested regeneration plots (circles) and transect points (triangles).

Long-Term Beech Study Plots

In addition to the stand plots, additional plots were established for long-term observation. Eighty-seven resistant beech trees were selected and paired with a nearby susceptible tree, for 174 study trees total (Houston 2001). Trees were cut or left as pairs. Houston (2001) established discrete, 15-ft radius plots around each of these trees, and 20 sprouts or seedlings were selected and tagged for long-term study. We located 171 of the original 174 study plots. Sprout maps from the original study were used to locate the original 20 tagged sprouts and seedlings. It was recorded whether or not the original sprout or seedling was alive or dead, and the maps were updated to reflect any mortality found. Sprouts or seedlings were retagged and recorded separately. Within the 15-ft radius plot, the total number of sprouts and seedlings within the plot were counted and recorded.

Statistical Analysis

Mortality data acquired in the 15-ft radius study plots were pooled within blocks to determine the number of living stems, calculated by dividing the total number of tagged sprouts found alive by the total number of sprouts that were there originally. A three-way analysis of variance (ANOVA) was used in conjunction with a general linear model to determine differences among main effects: harvest treatment, resistance/susceptibility of study tree, and cut/leave status of study tree against the response variable percentage alive. Due to lost degrees of freedom because of the lack of cut trees in the control blocks, a general linear model was run with the same data, and hypothesis tests were done on each factor level. Two other three-way ANOVAs were examined to determine whether there were main effects associated with season of harvest and type of cut (clearcut or partial cut). Pearson's correlation was used to test for normality, and a Levine's test was done to test for constant variance in all cases. Standing study trees from the long-term treatment plots were separated by treatment block and their resistant or susceptible status, and a percentage survival per treatment was determined.

Results

Overstory Composition

The 15 10-ac blocks differed in overstory composition and structure. The mean basal area per acre by species for each of the treatment blocks was calculated from the data collected along the transect points in each of the treatment blocks. Beech was most abundant in all treatments, being less prevalent in the summer and winter clearcuts (Table 1). Mean block basal area ranged from 50 to 128 ft^2/ac. Most of the blocks had small amounts of yellow birch, striped maple, and sugar maple. Softwoods and other species of hardwoods were present throughout the treatments at low levels, with the exception of the summer clearcut treatment, which had 38% of the basal area of other hardwoods.

Table 1. Mean basal area (ft^2/ac.) for species by treatment.[a,b] Empty values indicate species not present.

Species	WC		WP		SP		SC		NC	
American beech	33	(9.8)	94	(4.6)	67	(3.5)	24	(5.4)	81	(5.4)
Yellow birch	17	(5.5)	5	(0.67)	15	(2.7)	9	(1.4)	13	(1.7)
Striped maple	13	(2.6)	7	(1.0)	16	(3.2)	7	(1.5)	6	(1.5)
Sugar maple	8	(5.0)	4	(0.89)	4	(0.58)	3	(5.8)	18	(3.0)
Other hardwoods	5				12		38		4	
Softwoods			5		13		3		8	
Total	76		110		114		81		122	
Percent Beech	49%		85%		59%		30%		66%	

[a] Standard errors are in parentheses.
[b] WC, winter clearcut; WP, winter partial cut; SP, summer partial cut; SC, summer clearcut; NC, no cut (control).

Regeneration Composition

Beech was present in the regeneration in all blocks (Table 2) but was not most abundant in all blocks, nor was it equally prevalent in all treatments. In the 3.28-ft radius plots in which smaller regeneration was tallied, summer clearcuts were composed of trembling aspen, sugar maple, hobblebush, balsam fir, and yellow birch, with beech making up only a minor component. Beech was also a minor component of the regeneration in the winter clearcuts. No trembling aspen, big-toothed aspen, or balsam fir were present. Summer partial cuts were also dominated by beech, with striped maple, sugar maple, and yellow birch making up a large portion of the regeneration. Winter partial cuts overall were dominated by beech, with hobblebush and sugar maple composing much of the understory. Regeneration in the uncut blocks, although primarily composed of beech, had a strong presence of sugar maple, yellow birch and white ash.

The 6.56-ft radius plots were used to assess older and larger regeneration. Beech was not the most abundant in this size class in any treatment (Table 2). Striped maple was present in all treatments, as was beech and yellow birch. There was a high proportion of balsam fir in the no-cut control blocks.

Beech Regeneration Survival, 10 Years Post-Harvest

The general linear model using harvest treatment, resistance/susceptibility of study tree, and cut/leave status of study tree against the response variable percentage alive met the assumptions of constant variance ($P = 0.04$). Survival had a significant response to harvest treatment ($P = 0.01$) but not with other main effects or interactions. Average overall survival of beech regeneration in the summer harvests was 29% (23% clearcut, 35% partial cut), and the mean survival in winter harvests was 44% (40% clearcut, 49% partial cut) (Figure 2). The three-way ANOVA using season of harvest, resistance/susceptibility of study tree, cut/leave status of study tree against percent survival ($\alpha = 0.05$) was tested for normality using Pearson's correlation ($r = 0.987$), and a Levine's test was used to test for constant variance. Main effects were found with season of harvest ($P = 0.001$). Survival was lowest in summer harvest treatments. No significant interactions were found.

Average overall beech survival in clearcuts was 31% (summer, 23%; winter, 40%), whereas the average survival in partial cuts was 42% (summer, 35%; winter, 49%). The three-way ANOVA using type of cut (partial cut or clearcut), resistance/susceptibility of study tree, cut/leave status of study tree against percent survival ($\alpha = 0.05$) was tested for normality using Pearson's correlation ($r = 0.990$). Constant variance was tested using the Levine's test. Main effects were found associated with type of cut ($P = 0.029$). Survival was lowest in clearcut treatments. No significant interactions or other main effects were found.

Table 2. Regeneration in thousands of stems per acre in 3.28-ft[a] and 6.56-ft[b] radius plots. Empty values indicate species not present.

Species	NC[c] 3.28 ft	NC[c] 6.56 ft	SC 3.28 ft	SC 6.56 ft	SP 3.28 ft	SP 6.56 ft	WC 3.28 ft	WC 6.56 ft	WP 3.28 ft	WP 6.56 ft
American beech	12.4	0.8	3.4	0.9	11.0	1.0	4.3	0.7	18.4	0.8
Balsam fir	3.4	7.1	12.9	2.5	4.3	1.4	0.6		1.3	
Big tooth aspen			2.1	0.3						
Hobblebush	7.4		10.4		3.6	0.6	5.2	1.3	12.6	
Hophornbeam	2.6	0.3			2.6		1.6		2.6	
Red maple	11.5		2.9	2.8	7.9		5.3	0.9	3.6	
Red spruce	1.3			0.3	2.9		1.9		1.1	
Striped maple	4.8	1.2	2.9	1.4	8.2	1.2	2.9	1.2	2.9	0.8
Sugar maple	10.9		5.9	2.1	8.9		11.7	0.8	10.1	
Trembling aspen			9.5	1.7						
White ash	11.2		3.0	1.9	1.3		2.2	1.3	1.3	
Yellow birch	12.0		7.3	1.7	7.8		3.9	0.9	5.6	0.3

[a] One-m plots include all stems <1 in. in diameter and <39.37 in. tall.
[b] Two-m plots include all stems >1 in. in diameter and <5 in. in diameter and >39.37 in. tall.
[c] WC, winter clearcut; WP, winter partial cut; SP, summer partial cut; SC, summer clearcut; NC, no cut (control).

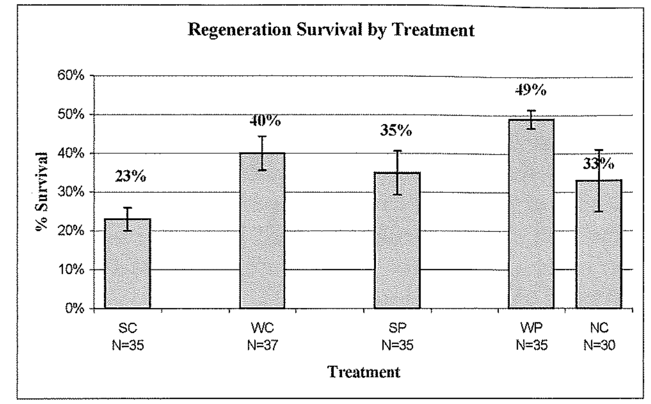

Figure 2. American beech regeneration survival by treatment.

Study Tree Survival

Survival data on study trees left standing were pooled by treatment to detect differences in mortality among resistant trees. Resistant study trees suffered 78% mortality in summer clearcut blocks and 67% mortality in winter clearcuts (Figure 3). Diseased trees all suffered some level of mortality, whereas resistant trees suffered no mortality in summer or winter partial cuts or in the control blocks.

Discussion

Beech Regeneration Survival in Summer versus Winter Harvests

The analysis of the survival of beech regeneration shows that harvest treatment affected regeneration survival. Sprout initiation is influenced by exposure to light and to higher temperatures (Maini and Horton 1966, Held 1983, Jones and Raynal 1986). The light level available to regeneration is considered to be the most important influence on growth rates in northeastern forests (Lorimer 1981, Runkle 1984), although a link between light level and long-term survival has not been studied. This study considered regeneration survival 10 years post-harvest. There was little difference in survival between winter clearcut (40%) and winter partial-cut blocks (49%). However, summer harvest blocks had lower survival, with clearcuts (23%) having less survival than partial cuts (35%). Jones and Raynal (1988) found that callus development is affected by season and by root injury, but exposed roots had limited callus formation and low survival rates. In the summer clearcut blocks, in particular, many of the parent roots were visible aboveground.

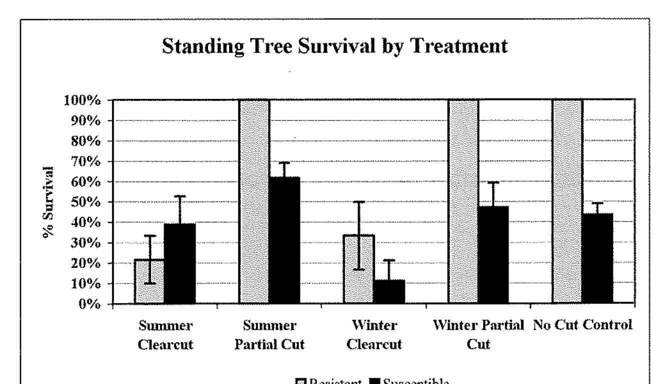

Figure 3. American beech standing tree survival by treatment.

Jones and Raynal (1988) found that fewer sprouts were initiated when roots of uncut beech were wounded in the fall than in the spring. They attributed this to the high mortality of fall-wounded roots, which have a less vigorous callus formation (Jones and Raynal 1988). Houston's (2001) measurements taken shortly after harvest indicated that there were more sprouts in the summer cuts than the winter cuts, but that after 3 years of measurements, the high mortality found in the summer cuts negated any difference found. We have found that the mortality of summer regeneration continued and that by 10 years after harvest, beech regeneration survival in the summer treatments (29%) was lower than in the winter treatments (41%). Survival of regeneration ranged from a low of 23% around cut trees in summer clearcuts to a high of 49% around cut trees in winter partial cuts (Figure 2). Nutrient levels in roots have been found to be highest in the late fall and early winter, which may increase the survival of sprouts that depend on the parent tree for nutrients (Taiz and Zeiger 2002).

Houston (2001) found that the lowest regeneration mortality was in the no-cut control blocks, where the regeneration was mostly composed of seedlings. Interestingly, 10 years post-harvest, regeneration survival was higher in the winter partial-cut blocks (49%), summer partial-cut blocks (35%), and winter clearcut blocks (39%) than in the control blocks (33%). Seedlings are known to have a higher mortality than sprouts in some cases (Ward 1961). The oystershell scale, *Lepidosaphes ulmi* L., was responsible for some mortality in all blocks when the study was initiated. Indeed, Houston (2001; David R. Houston, USDA Forest Service (retired), personal communication, Dec. 13, 2002) stated that it was the most important mortality agent at that time, although he indicated no apparent treatment effects.

The survival of the sprouts in the plots was not directly related to whether trees were cut or left standing. Overall, there was not a significant difference in survival between monitored sprouts around cut trees (35% survival) than around uncut trees, (37% survival). This finding is different from what Houston (2001) found 3 years after harvest. He found that significantly more sprouts died around cut trees. This may have been a consequence of the high mortality found in the summer clearcut treatments, where there was a greater disturbance.

Regeneration Survival of Resistant and Susceptible Trees

There was no significant relationship between the mortality of the monitored regeneration and the resistance/susceptibility status of the study plot tree. Overall, 39% of the regeneration around resistant, high-vigor trees survived compared to 34% for those around susceptible, low-vigor trees. Likewise, there was no difference in regeneration survival found between sprouts from resistant trees that were cut (39%) and those that were not cut (39%). There was little difference in survivorship of regeneration around susceptible trees that were cut (32%) and those that were left standing (35%).

Standing Tree Survival

The fate of resistant beech left standing in the various cuts is important to consider. These trees, which show resistance to the beech scale, are the only reliable hard mast producers in many areas and are vital to the future of the species. Past management recommendations for landowners conducting harvesting in stands affected by beech bark disease have been to identify and retain resistant trees, even in clearcuts (Houston 2001). The results of this study show that in clearcut blocks, resistant trees left standing suffered mortality ranging from 67 to 78%. We found no mortality of resistant trees left standing in any of the partially cut blocks or in the no-cut control blocks. This is a significant result, because 10 years ago, most of these trees were listed as high-vigor trees (Houston 2002). We found that the majority of the resistant trees in the clearcut blocks were standing dead with sunscald symptoms. These trees had not been dead for a long time. We suggest that the probable causes of this mortality are exposure and sunscald. Other resistant trees in the clearcuts that were not dead were declining and had very little live crown. Not a single resistant tree died in any of the other treatment blocks.

As part of Houston's (2001) study, both resistant and susceptible trees were left standing in all treatments. It is not surprising that we found that many of the standing susceptible trees had died in clearcuts, as they were suffering from the effects of beech bark disease. Mortality was common in every treatment block, including the control blocks. The mortality of susceptible trees left standing was the highest in clearcut blocks, averaging 61% in the summer clearcut blocks and 89% in the winter clearcut blocks. In summer clearcut blocks, a higher percentage of the resistant trees died (78%) than did susceptible trees (61%).

Management Implications

Many stands have been rendered less productive as a result of a high level of defective beech and dense thickets of beech reproduction. We found no significant difference between the harvest-initiated sprouting around resistant and susceptible trees. Because American beech is so shade-tolerant, it does not experience rapid mortality when regeneration is left to grow in the understory. It is in this way that beech is often able to work its way into the overstory by taking advantage of gaps that occur over time and by outliving other hardwood species (Lorimer 1981). We have found that beech will experience greater mortality if regeneration is exposed in clearcuts.

Winter partial cuts, although they initially cause the least sprouting overall, resulted in the highest sprout survival 10 years following the harvesting treatments. The decreased sprouting in the winter cut blocks is thought to be the result of the root systems being protected by frozen soil or by snow cover and thus less likely to suffer injury as a result of harvesting activities (Houston 2001).

Leaving resistant trees and removing susceptible ones should enhance genetic variation through the sexual reproduction of resistant trees. In addition, large diameter, mature beech have a high wildlife value for the mast they produce. Past studies have suggested leaving resistant trees in all harvesting situations. However, this study shows that leaving resistant American beech of high vigor in clearcuts without the protection of surrounding trees leaves them susceptible to decline and death from exposure. When implementing management plans, resistant trees should be identified and retained, but care should be taken to ensure that they are protected by the presence of surrounding trees.

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Short-Term Effects of Silvicultural Treatments on Microsite Heterogeneity and Plant Diversity in Mature Tennessee Oak-Hickory Forests

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ABSTRACT: *Growing emphasis on sustainability has increased the demand for information on effects of forest management on species diversity. We investigated the hypothesis that plant diversity is a function of microsite heterogeneity by documenting plant diversity and heterogeneity in canopy cover, light, and soil moisture produced by four silvicultural treatments during the first growing season following treatment: prescribed burning, wildlife retention cut with prescribed burning, wildlife retention cut, and shelterwood cutting. Treatments and controls were randomly assigned within four relatively undisturbed, 70–90-year-old oak-hickory stands. Heterogeneity in canopy cover and photosynthetically active radiation was greatest after shelterwood cutting, whereas the wildlife retention cut resulted in less removal of canopy trees and a smaller increase in heterogeneity of these factors. The addition of prescribed burning enhanced the effects of the wildlife retention cut. Prescribed burning alone had the least impact on heterogeneity of these factors. Soil moisture variability appeared to be independent of treatments. Shelterwood cutting increased first-year herbaceous plant diversity, and this increase was likely due, in part, to increased heterogeneity in canopy cover, light, and seedbed condition. These first-year results partially support the hypothesis that plant diversity is a function of microsite diversity in these forests. Long-term monitoring is underway. North. J. Appl. For. 23(3):197–203.*

Key Words: Heterogeneity, silviculture, photosynthetically active radiation, diversity, Tennessee.

Emphasis on sustaining and restoring species diversity has increased in recent years because of the adoption of an ecosystem management paradigm by public management agencies and the implementation of certification programs for managing both public and private forests. These new priorities have increased the demand for information on the effects of different management practices on the diversity of commercial and noncommercial species alike. Higher levels of plant and animal species diversity can protect ecosystem communities from invasion by nonnatives and can reduce the impact and likelihood of disease outbreaks (Knops et al. 1999). Diverse communities of plant species can increase productivity and increase the ability of a plant community to sustain itself (Tilman et al. 1996). Insect diversity and animal population stability have also been shown to increase with increasing levels of habitat or microsite diversity (Michael 1995, Knops et al. 1999).

It has been hypothesized that the diversity of plants and animals in an ecosystem is linked closely to heterogeneity in multiple site factors, such as soil type, moisture regime, and light availability, although the accumulation of empirical evidence for this is ongoing (Naeem and Colwell 1991, Barnes et al. 1998). In addition to the effects of landform, parent material, and soil type, heterogeneity in site factors is strongly influenced by the stage of stand development,

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